

Machine-Checked Status of the SG–Cole–Hopf Route to 3D Navier–Stokes Global Regularity: H1 Obstruction, Stochastic-Lagrangian Repair, and Pancake Closure

The MeTTapedia formalization effort

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Abstract

Ben Goertzel’s manuscript *Global Regularity of 3D Navier–Stokes via SG–Cole–Hopf Program* (v7) reduces global regularity of the 3D incompressible Navier–Stokes equation on $\mathbb{T}^3$ to a uniform-in-time identity-energy bound on the volume-preserving diffeomorphism group $\text{SG} = \text{SDiff}(\mathbb{T}^3)$. The June 2026 MeTTapedia formalization found that the route’s original local adjoint hypothesis (H1) is false in a normalized Fourier-mode shear model: inside every positive H^m chart radius there are area-preserving shears of fixed H^m displacement size whose adjoint $H^m \rightarrow H^m$ ratio is $\sqrt{1 + (\varepsilon k)^2}$ and therefore unbounded. That refutation remains intact.

This September revision records the follow-up repair campaign in full. The corrected route does not rely on the false $H^m \rightarrow H^m$ adjoint bound; it replaces it with a stochastic-Lagrangian/cotangent transport skeleton and a microlocal vorticity-stretching gate. The machine-checked surface now comprises: the evasion contrast showing the H1-breaking shear family leaves pointwise Jacobians untouched; the one-form transport calculus with the exact frame-Laplacian identity; the divergence-free non-exact witness that forces the gauge; the stochastic-Lagrangian pushdown deriving Navier–Stokes with a fully explicit pressure; the conditional BKM chain with its gate consumers; nonzero exact inhabitants and end-to-end canaries; the frozen-strain pancake obstruction with its exact peak-amplification law; the plane-wave Biot–Savart null structure claimed in the pancake-closure proposal; and the dyadic absorption step that would turn the pancake-sector estimate into a common BKM envelope. The September field-transfer work additionally constructs the coherent/misaligned projection for arbitrary complex periodic Fourier coefficients, transfers the coherent stretching estimate to those coefficients, proves the exact square-dyadic cone lattice count, and verifies the anisotropic L^1 kernel mechanism that preserves the receiver’s L^∞ norm. It also proves the estimate at continuous real frequencies and for arbitrary projected amplitudes, constructs the exact periodic operator-kernel realization on finite Fourier fields, transports the cancellation to every frozen oriented strain frame, and now transports the full square-dyadic periodized kernel and its cardinality-free C/N endpoint uniformly to those frames. Complexifying the frame and conjugating both inputs and the output gives a physical-frame bound; uniformity then permits an arbitrary frame choice at each output point without a frame-derivative loss at this L^∞ kernel endpoint. A separate exact variable/frozen split and a spectral-gap perturbation theorem remain available for formulations that differentiate the symbol. The physical finite-sector stretching sum is now proved to equal this coherent annular kernel action plus the exact three-term misalignment remainder, both in an arbitrary frozen frame and in the full pointwise frame constructed from the strain. This frame maps the most expanding eigenvector to the coherent reference e_x and the most compressive eigenvector to the axial reference e_z . Hence the coherent term has a cardinality-free C/N bound without an abstract split hypothesis. A further checked projection argument bounds the exact complex Fourier misalignment coefficient by its energy transverse to the expanding line.

This estimate sums over the genuine finite pancake sector without a mode-count loss, and finite Parseval identifies the right-hand side with a physical-space L^2 transverse-energy integral for a frozen full strain frame. A sign-invariant line-projector perturbation theorem further controls freezing error by $4\varepsilon/\text{gap}$ times the full L^2 energy, with an explicit small frozen-top-gap alternative. On the separated branch, a homogeneous spectral-gap estimate now converts the frozen transverse energy into the integrated top Rayleigh defect, without normalizing vorticity at its zeros. Finite aggregation of the coercive cells and generic hard-set/smooth-weight bounded-overlap integral estimates are also checked. The concrete cover now permits a different frozen strain on each of its 125 cells: on the separated branch the complete arbitrary-data misalignment energy is bounded by the sum of localized frozen Rayleigh defects plus one fixed-cover projector-oscillation error. The sign-invariant expanding-line projector is quantitatively locally Lipschitz wherever the top eigenvalue is simple, so a continuous strain field gives a continuous, measurable transverse-energy density on every separated-gap region without choosing an eigenvector sign. The top-gap collision band is further resolved by the stable complementary cluster: if the bottom gap remains open, the expanding two-plane projector is locally Lipschitz and measurable; if both adjacent gaps are small, trace-free strain forces the whole spectrum to be small and yields a direct small-rate stretching bound. On the stable-plane branch, the sign-invariant energy in the contracting line now satisfies an exact direction-evolution law and a bottom-gap-weighted residence inequality. Its only undischarged forcing channels are the non-strain direction forcing and motion of the contracting line. The direction evolution has now also been attached to the exact hard square-dyadic pancake block. Coefficientwise time differentiability gives the termwise derivative of its finite Fourier reconstruction, coordinatewise real part is realized as a continuous real-linear map, and a nonzero factorization of that same block obeys the exact strain-plus-residual direction law. The residual is displayed explicitly as the reconstructed block derivative minus pure strain stretching; transport, viscosity, interscale transfer, and localization commutators must be controlled there. This removes the invalid shortcut of using coherence of the total vorticity direction as if it automatically controlled the projected block. The pair-indexed phase-suppressed Biot–Savart stretching polynomial is now also collected over equal output frequencies into a canonical coefficient field. A subsequent unit-torus normalization restores the Fourier phases: curl inversion contributes $-(2\pi i)^{-1}$ and differentiation contributes $2\pi i$, so the physical strain–vorticity coefficient is the *negative* of the phase-suppressed symbol. This sign identity, the right-hand-side transport coefficient, viscosity, and the resulting finite vorticity RHS are all checked before they enter the two-channel block residual. Finally, the hard shell-and-cone selector is proved invariant under $k \mapsto -k$. Hence physical conjugate symmetry $\widehat{\omega}(-k) = \widehat{\omega}(k)$ makes the selected complex reconstruction exactly real-valued. The real block in the direction law is therefore the actual physical pancake block, not a lossy projection. The unit-torus Biot–Savart, strain, transport, and viscous symbols are furthermore proved to preserve conjugate symmetry, and a finite-pair reindexing theorem shows that the complete finite vorticity RHS preserves the real Fourier subspace. The frequency-support part of the analytic localization bridge is now closed: for fixed normalized cutoff margins, a four-chart multiplier is constructed which is exactly coherent on every leaked output mode and has a uniform inverse-kernel L^1 bound. For finite-band cutoffs this multiplier is now transferred through square-dyadic scaling, rotation, periodization, and physical frames, and applied exactly to the localized products. The concrete localization layer is now closed as well: a compactly supported smooth bump is periodized, its Fourier coefficients are proved absolutely summable, its 125 translated three-torus cells admit one common finite approximation, and the actual smooth-cell interaction differs from the coherent finite convolution by an explicit $O(\varepsilon/N)$ error. Complete ordinary dyadic shells are also constructed for arbitrary Fourier data, split exactly into pancake and non-pancake shares, and reconciled with the square-dyadic multiplier scales. For componentwise L^2 fields their shell energies form a summable series; the zero-mode energy plus that series is exactly the physical L^2 energy. The remaining smooth projector collar, space–time treatment of the small-gap branch, and conversion from this L^2 /defect control to the scale-local endpoint budget remain open. The latest audit also isolates why the proposed pancake closure does not yet prove the Millennium statement: its high-low split covers all aperture angles iff the scale-local strain

satisfies $\gamma \leq \nu \lambda_q$, exactly the quantitative budget it set out to prove. The surviving open input is the leading route-specific pointwise-in-time Lean predicate `ScaleLocalPancakeStrainBudget`, a sharpened version of the earlier `StochasticStretchingEstimate`. Identification of the new hard dyadic blocks with the route’s dynamical misalignment functional, the stated non-pancake sector estimates, the smooth projector passage, and the final BKM continuation plumbing are additional explicit obligations. Thus the status is neither a completed Navier–Stokes proof nor a vague “hard analytic gap”: it is a conditional route with a named, scale-local, machine-audited leading analytic pin and a finite list of surrounding formalization obligations.

1 The problem and the role of hypothesis (H1)

1.1 The SG–Cole–Hopf route

We work on the flat three-torus $\mathbb{T}^3$ with smooth divergence-free initial data u_0 . Ben Goertzel’s manuscript *Global Regularity of 3D Navier–Stokes via SG–Cole–Hopf Program*, version v7 (hereafter “v7”), lifts the incompressible Navier–Stokes equation to a diffusion on the volume-preserving diffeomorphism group

$$\text{SG} := \text{SDiff}(\mathbb{T}^3),$$

equipped with a right-invariant Fourier frame $\{D_i\}_{i \geq 1}$ induced by a divergence-free frame $\{E_i\}_{i \geq 1}$ on $\mathbb{T}^3$. The generator and carré du champ are

$$L := \sum_{i \geq 1} D_i^2, \quad \Gamma(F) := \sum_{i \geq 1} (D_i F)^2.$$

One solves the SG heat equation $\partial_t \Phi = \nu L \Phi$ with a strictly positive datum $\varphi = \exp(-W_0/(2\nu))$, applies the Cole–Hopf transform $W = -2\nu \log \Phi$ to obtain an SG Hamilton–Jacobi–Bellman potential, and pushes the Γ -gradient down at the identity to a projected Navier–Stokes solution $u(t)$ on $\mathbb{T}^3$.

The analytic content is concentrated entirely in closing the Beale–Kato–Majda (BKM) vorticity criterion. With $u(t) = \text{Ev}(W(t))$ and $\omega(t) = \text{curl } u(t)$, the push-down identity and Cauchy–Schwarz give a frame constant $C_{\text{curl}E} < \infty$ with

$$\|\omega(t)\|_{L^\infty(\mathbb{T}^3)} \leq \sqrt{C_{\text{curl}E}} \sqrt{\Gamma(W(t))(\text{id})}, \quad (1)$$

so that a uniform-in-time bound on $\Gamma(W(t))(\text{id})$ makes the BKM integral $\int_0^T \|\omega(t)\|_{L^\infty} dt$ finite on every finite interval. By the log chain rule $\Gamma(\log \Phi) = \Gamma(\Phi)/\Phi^2$ and the maximum principle $\Phi(t, \text{id}) \geq m_\varphi$ (v7, Prop. 7.1),

$$\Gamma(W(t))(\text{id}) = 4\nu^2 \Gamma(\log \Phi(t))(\text{id}) \leq \frac{4\nu^2}{m_\varphi^2} \Gamma(P_t \varphi)(\text{id}), \quad (2)$$

which reduces the whole program to a single uniform bound on $\Gamma(P_t \varphi)(\text{id})$.

1.2 Reduction to a local energy bound, and hypothesis (H1)

Rather than invoking a global Bakry–Émery Γ_2 curvature lower bound, v7 obtains that single bound from a *local* package. Let $\kappa : U \rightarrow V \subset H_\sigma^m(\mathbb{T}^3)$ be the chart of a small SDiff^m neighborhood of the identity ($m \geq 5$ integer, with the Γ -Hilbert structure at id equivalent to the H_σ^m norm), let χ be the cutoff defining W_0 and φ , and set the cutoff region

$$K := \kappa^{-1}(\text{supp } \chi) \subset U.$$

Outside K one has $W_0 \equiv 0$, hence $\varphi \equiv 1$ and $\Gamma(\varphi) \equiv 0$. On K , v7 assumes two local hypotheses (v7, §5.3 and §6, displayed at Lemma D.18):

$$(H1) \text{ Adjoint bounded on support: } \exists C_{\text{Ad}} < \infty : \|\text{Ad}_g\|_{H^m \rightarrow H^m} \leq C_{\text{Ad}} \quad \text{for all } g \in K, \quad (3)$$

$$(H2) \text{ Finite cutoff energy: } \exists C_\varphi < \infty : \Gamma(\varphi)(g) \leq C_\varphi^2 \quad \text{for all } g \in K. \quad (4)$$

From (3)–(4) v7 derives the key estimate via a conjugation differentiation identity (v7, Prop. 6.5, proved as Lemma F.7): for all $t \geq 0$,

$$\Gamma(P_t \varphi)(\text{id}) \leq C_{\text{Ad}}^2 C_\varphi^2. \quad (5)$$

The mechanism is transparent. In the flow representation $P_t F(g) = \mathbb{E}[F(G_t \circ g)]$, differentiating along a conjugation curve $C_{t,i}(\varepsilon) = g \circ \exp(\varepsilon E_i) \circ g^{-1}$ produces the direction $\text{Ad}_g E_i$, and Cauchy–Schwarz in the Γ -Hilbert structure gives the pointwise bound

$$|D_{\text{Ad}_g E_i} \varphi(g)|^2 \leq \|\text{Ad}_g\|_{\text{op}}^2 \Gamma(\varphi)(g) \leq C_{\text{Ad}}^2 C_\varphi^2 \mathbf{1}_{\{g \in K\}}, \quad (6)$$

after which a Jensen/energy reduction yields (5). Chaining (5) through (2) and (1) closes BKM. Hypothesis (H1) is therefore the single load-bearing analytic input of the route: *every* subsequent estimate inherits the constant C_{Ad} from (3), and the program establishes global regularity if and only if (3) holds.

1.3 How v7 attempts to verify (H1)

v7 verifies (H1) at Lemma 6.1 (and again at Lemma D.18). The adjoint / push-forward action is written, for $g \in \text{SDiff}^m$ and $v \in H_\sigma^m$, as

$$\text{Ad}_g v := (Dg)(v \circ g^{-1}), \quad (7)$$

and the argument is that, for $m > 5/2$, composition and inversion are continuous (indeed smooth) operations on SDiff^m , so the map $(g, v) \mapsto \text{Ad}_g v$ is continuous from $\text{SDiff}^m \times H_\sigma^m$ into H_σ^m ; hence $g \mapsto \|\text{Ad}_g\|_{\text{op}; H^m \rightarrow H^m}$ is locally bounded near id , and one shrinks K until (3) holds.

The gap is in (7) itself. The factor Dg is the Jacobian of g , and a map $g \in H^m$ has Jacobian $Dg \in H^{m-1}$ only. The adjoint thus carries a *derivative loss*: Ad_g maps H^m into H^{m-1} , and there is no uniform $H^m \rightarrow H^m$ operator bound near the identity in the H^m topology. This is the classical Ebin–Marsden phenomenon: right translation on SDiff^m is smooth, but the adjoint of the right-invariant frame is not a bounded H^m operator uniformly over an H^m -neighborhood. The local-boundedness argument conflates topological-group continuity of composition with the operator-norm estimate (3) that the route actually needs.

2 The Fourier-mode shear model and the checked $\neg(\text{H1})$ theorems

We make the derivative-loss obstruction precise and machine-checkable in a clean analytic model, on the normalized two-torus, of an area-preserving shear family

$$g_k(x, y) = (x, y + a_k \sin(kx)), \quad \delta_k := g_k - \text{id} = (0, a_k \sin(kx)).$$

The amplitude a_k is chosen so that the displacement has *fixed* H^m size ε for every wavenumber k ; thus the whole family sits inside any prescribed chart radius. Applying the push-forward (7) to the constant horizontal field $v = (1, 0)$ gives

$$\text{Ad}_{g_k} v = (1, a_k k \cos(kx)),$$

and, with the same H^m normalization, the operator ratio is

$$\frac{\|Ad_{g_k} v\|_{H^m}}{\|v\|_{H^m}} = \sqrt{1 + (\varepsilon k)^2}.$$

The displacement stays at size ε while the ratio grows without bound: no finite C_{Ad} can dominate it on any positive chart radius.

This is formalized, with no sorry and by elementary means, in the Lean module `Mettapedia.FluidDynamics.NavierStokes.BenH1Break`. The model quantities are defined as follows.

```

/-- In the normalized two-torus shear family, every selected displacement has
fixed 'H^m' size 'epsilon'. -/
def benH1ModeDisplacementHNorm (epsilon : Real) (_k : Nat) : Real :=
  epsilon

/-- Lower-bound ratio obtained by applying the shear push-forward to the
constant horizontal vector field. -/
def benH1ModeAdjointRatio (epsilon : Real) (k : Nat) : Real :=
  Real.sqrt (1 + (epsilon * (k : Real)) ^ 2)

/-- The linearized commutator part alone grows like 'epsilon * k'. -/
def benH1ModeCommutatorRatio (epsilon : Real) (k : Nat) : Real :=
  epsilon * (k : Real)

```

Hypothesis (H1), restricted to this normalized family inside a chart radius ρ (taking $\varepsilon = \rho/2$ so the family is strictly inside the ball), is the predicate:

```

/-- H1 restricted to the normalized Fourier-mode shear family inside chart
radius 'rho'. -/
def benH1FourierModeUniformBoundWithinRadius (rho C : Real) : Prop :=
  forall k : Nat,
    benH1ModeDisplacementHNorm (rho / 2) k <= rho ->
      benH1ModeAdjointRatio (rho / 2) k <= C

```

The core fact is that the adjoint ratio dominates the linear commutator part εk , which is unbounded over k . The proof is fully elementary: it picks $k > C/\varepsilon$, lower-bounds $\sqrt{1 + (\varepsilon k)^2}$ by εk , and derives a contradiction.

```

theorem benH1ModeAdjointRatio_unbounded {epsilon C : Real}
  (hepsilon : 0 < epsilon) :
  Not (forall k : Nat, benH1ModeAdjointRatio epsilon k <= C) := by
  intro hbound
  obtain ⟨k, hk⟩ := exists_nat_gt (C / epsilon)
  have hC_lt : C < epsilon * (k : Real) := by
    have hmul := mul_lt_mul_of_pos_left hk hepsilon
    field_simp [ne_of_gt hepsilon] at hmul
    exact hmul
  have hnonneg : 0 <= epsilon * (k : Real) := by positivity
  have hlinear_le_ratio :
    epsilon * (k : Real) <= benH1ModeAdjointRatio epsilon k := by
    unfold benH1ModeAdjointRatio
  have hsquare :
    (epsilon * (k : Real)) ^ 2 <= 1 + (epsilon * (k : Real)) ^ 2 := by

```

```

    linarith
  calc
    epsilon * (k : Real)
      = Real.sqrt ((epsilon * (k : Real)) ^ 2) := by
        rw [Real.sqrt_sq_eq_abs, abs_of_nonneg hnonneg]
    _ <= Real.sqrt (1 + (epsilon * (k : Real)) ^ 2) :=
      Real.sqrt_le_sqrt hsquare
  exact not_lt_of_ge (le_trans hlinear_le_ratio (hbound k)) hC_lt

```

From this the refutation of (H1) on the model follows directly. Any positive chart radius ρ admits the normalized family at $\varepsilon = \rho/2$ (whose displacements all have H^m size $\rho/2 \leq \rho$), and its adjoint ratios are unbounded, contradicting any proposed finite C .

```

/-- Any positive chart radius contains the normalized shear family with
'epsilon = rho / 2', and its adjoint ratios are unbounded. -/
theorem not_benH1FourierModeUniformBoundWithinRadius {rho C : Real}
  (hrho : 0 < rho) :
  Not (benH1FourierModeUniformBoundWithinRadius rho C) := by
  intro hbound
  have hepsilon : 0 < rho / 2 := by positivity
  have hmode_bound : forall k : Nat, benH1ModeAdjointRatio (rho / 2) k <= C := by
    intro k
    apply hbound k
    unfold benH1ModeDisplacementHNorm
    linarith
  exact benH1ModeAdjointRatio_unbounded hepsilon hmode_bound

/-- Compact public verdict: Ben's H1, as a uniform 'H^m -> H^m' adjoint bound,
is false on the normalized Fourier-mode shear model for every positive chart
radius and every proposed finite constant. -/
theorem benGoertzelH1_false_in_fourierModeShearModel :
  forall rho C : Real,
    0 < rho -> Not (benH1FourierModeUniformBoundWithinRadius rho C) := by
  intro rho C hrho
  exact not_benH1FourierModeUniformBoundWithinRadius (rho := rho) (C := C) hrho

```

The three checked statements are `benH1ModeAdjointRatio_unbounded`, `not_benH1FourierModeUniformBoundWithinRadius`, and the compact verdict `benGoertzelH1_false_in_fourierModeShearModel`: for every positive chart radius and every proposed finite constant, the modeled (H1) uniform $H^m \rightarrow H^m$ adjoint bound fails. (A companion lemma `benH1ModeCommutatorRatio_unbounded` records that the linearized commutator part εk is already unbounded on its own.)

3 The numeric lab: the norms are derived, not assumed

The Lean development encodes the two normalized quantities $\|\delta_k\|_{H^m} = \varepsilon$ and $\|\text{Ad}_{g_k} v\|_{H^m} / \|v\|_{H^m} = \sqrt{1 + (\varepsilon k)^2}$ as definitions. To confirm that these are not circular — that the factor k in the adjoint ratio is a genuine consequence of the Sobolev-weight structure and the derivative in (7), rather than an assumption — a companion numeric lab (`ns_ben_h1_fourier_mode_lab.py`, in the MeTTapedia validation lab) *computes* the norms from first quantities. With H^m Fourier weight $w_k = (1 + k^2)^{m/2}$:

- the amplitude is set to $a_k = \sqrt{2}\varepsilon/w_k$, so the displacement norm is $\|\delta_k\|_{H^m} = |a_k| w_k/\sqrt{2} = \varepsilon$ (the $\sqrt{2}$ is the L^2 normalization of $\sin(kx)$);
- differentiating, $\partial_x(a_k \sin(kx)) = a_k k \cos(kx)$, whose H^m norm is $|a_k| k w_k/\sqrt{2} = \varepsilon k$. The weight w_k *cancels* the amplitude, leaving ε ; the surviving factor k is the one derivative gained by Dg_k ;
- the constant field has $\|v\|_{H^m} = 1$, so the adjoint norm is $\sqrt{1^2 + (\varepsilon k)^2}$ and the ratio is $\sqrt{1 + (\varepsilon k)^2}$.

This is exactly the Ebin–Marsden derivative loss of (7) made quantitative: $g_k \in H^m$ has $Dg_k \in H^{m-1}$, and the action of Dg_k multiplies the mode’s high-frequency content by its wavenumber. Table 1 lists the lab output for $\varepsilon = 0.01$, $m = 3$. The displacement norm stays fixed at ε across four decades of k while the adjoint ratio grows.

k	amplitude a_k	$\ \delta_k\ _{H^m}$	commutator εk	ratio $\sqrt{1 + (\varepsilon k)^2}$
8	2.699×10^{-5}	0.0100	0.080	1.0032
16	3.433×10^{-6}	0.0100	0.160	1.0127
32	4.310×10^{-7}	0.0100	0.320	1.0500
64	5.393×10^{-8}	0.0100	0.640	1.1873
128	6.743×10^{-9}	0.0100	1.280	1.6243
256	8.429×10^{-10}	0.0100	2.560	2.7484
512	1.054×10^{-10}	0.0100	5.120	5.2167
1024	1.317×10^{-11}	0.0100	10.240	10.2887

Table 1: Fourier-mode lab output ($\varepsilon = 0.01$, $m = 3$). The displacement H^m norm is pinned at ε for every wavenumber, while the adjoint operator ratio $\sqrt{1 + (\varepsilon k)^2}$ grows without bound. For any proposed constant C the ratio first exceeds it near $k \approx \sqrt{C^2 - 1}/\varepsilon$ (e.g. $C = 2$ at $k = 174$, $C = 10$ at $k = 995$, $C = 100$ at $k = 10000$): no finite C_{Ad} survives.

4 The repaired stochastic-Lagrangian route: the full formalized layer

After the H1 obstruction was isolated, the formalization effort tested a different repair strategy. The repaired route abandons the false $H^m \rightarrow H^m$ adjoint estimate and moves the load-bearing estimate to the stochastic-Lagrangian derivative flow of Constantin–Iyer (*Comm. Pure Appl. Math.* **61** (2008), 330–345): the state is a momentum one-form $\mu = u^\flat + dW$ transported along a noisy flow, the velocity is recovered by averaging and projection, and the continuation estimate is a vorticity-stretching gate feeding the BKM criterion. In the Lean development the gate appears first as the coarse input `StochasticStretchingEstimate`, and in the dyadic refinement as the scale-local predicate `ScaleLocalPancakeStrainBudget`.

This section records the *whole* machine-checked surface of the repair campaign, layer by layer, not only the current crux. All Lean listings below are structurally verbatim from the sources (declaration names, hypotheses, and conclusions exactly as checked), with Unicode notation transliterated to ASCII for typesetting; the Lean sources are normative. Table 2 summarizes the layers.

Layer (module)	Headline results	Scope label
H1 model (<code>BenH1Break</code>)	adjoint ratio unbounded; $\neg(\text{H1})$	model refutation
Evasion (<code>ShearJacobianBounds</code>)	same family: pointwise Jacobian $\leq \varepsilon$	model contrast
One-forms (<code>OneFormFrameCalculus</code>)	frame-Laplacian identity $\sum_i L_{E_i}^2 = \Delta$	exact identity
Exactness (<code>OneFormExactness</code>)	div-free smooth non-exact witness	exact witness
Gradients (<code>GradientOneForm</code>)	Cartan formula; self-transport identity	exact identities
Pushdown (<code>ConstantinIyerRepresentation</code>)	NS with explicit pressure from transport	conditional (interface)
Instances (<code>DecayingShearInstance</code> , regression)	nonzero exact inhabitant; rest canaries	exact instances
Gate (<code>VorticityStretchingGate</code>)	pointwise common envelope from named pin	conditional
Countermodel (<code>FrozenStrainModel</code>)	exact pancake solution; no uniform depletion	model refutation
Aperture (<code>ApertureCoverageDichotomy</code>)	dichotomy $\iff$ strain budget; gap failures	model equivalence
Null structure (<code>PlaneWaveSelfStrain</code>)	single-mode Biot–Savart null stretching	exact identity
Dyadic closure (<code>DyadicPancakeClosure</code>)	absorption algebra; sharpened pin; route	conditional
Coherence (<code>PancakeCoherentPairEstimate</code>)	pairwise aperture gain	exact inequality
Refined pin (<code>MisalignmentRefinedPin</code>)	misalignment budget implies pancake budget	conditional
Direction dynamics (<code>DirectionEvolutionTilting</code>)	direction laws; gap-free tilting no-go	exact/refutation

Table 2: The formalized surface of the repair campaign (all modules under `Mettapedia.FluidDynamics.NavierStokes`, September 2026). “Model” statements are proved in named analytic models; “exact” statements are unconditional identities or witnesses on $\mathbb{R}^3$ -coordinate space; “conditional” statements are machine-checked implications whose analytic inputs are named hypotheses.

4.1 Why a pointwise transport route can evade the H1 countermodel

The first question for any repair is whether the Section 2 countermodel already defeats it. It does not, and this is itself a theorem. On *the same* H^m -normalized shear family that refutes (H1), the module `ShearJacobianBounds` proves (`shearFamily_pointwise_bounded_while_sobolev_ratio_unbounded`):

1. every pointwise Jacobian deviation is bounded by ε uniformly in the frequency k (`shearFamily_jacobian_uniform_bound`: $\|Dg_k(x)v - v\| \leq \varepsilon\|v\|$ for all x, v); while
2. the $H^m \rightarrow H^m$ operator ratio $\sqrt{1 + (\varepsilon k)^2}$ of the same family admits no finite uniform bound (Section 2).

The high-frequency-shear derivative-loss mechanism therefore lives entirely in the Sobolev operator topology: the quantity feeding a pointwise gate $\int_0^T \mathbb{E}\|DX_t\|_\infty dt$ is not stressed at all by this family. The checked statement is deliberately labeled: it does not prove any pointwise gate — it proves this family cannot refute one. That is what licenses moving the route from adjoint bounds to Lagrangian transport.

4.2 The one-form layer: non-exact momenta and the frame Laplacian

The repaired state is a momentum one-form with a gauge, $\mu = u^\flat + dW$, not a scalar potential. Two checked facts pin down why this shape is forced and why the frame generator acts correctly on it.

Non-exact momenta are necessary. Exact one-forms are closed (`oneFormExtDerivAt_eq_zero_of_exact`), and there is a smooth, everywhere divergence-free field that is not exact — the transverse sine mode:

```

theorem exists_divergenceFree_smooth_not_exact_dim3 :
  exists u0 : EuclideanSpace Real (Fin 3) -> EuclideanSpace Real (Fin 3),
    ContDiff Real 2 u0 /\
    (forall x, coordDivergence u0 x = 0) /\
    Not (IsExactOneForm u0)

```

A scalar-potential (pure Cole–Hopf) ansatz therefore cannot carry a general divergence-free momentum: vorticity lives exactly in the non-exact part. This is the constructive face of the Section 1 diagnosis, and it is what the additive gauge dW is for.

The frame reproduces the Laplacian. The route’s generator is a sum of squared frame derivatives (v_i ’s $L = \sum_i D_i^2$). On the truncated trigonometric coordinate frame (three axes, transverse polarizations, both phases) the squared Lie derivatives of a one-form sum *exactly* to the coordinate Laplacian: with amplitude $1/\sqrt{2}$ in dimension 3,

$$\sum_j \sum_{p \neq j} \left(L_{\cos\text{-mode}(j,p)}^2 + L_{\sin\text{-mode}(j,p)}^2 \right) \mu = \Delta \mu$$

(`frameLaplacian_oneForm_normalized`, from the general $(\text{card } \iota - 1)a^2$ identity `frameLaplacian_oneForm` and the phase-pair cancellation `lieOneForm_sq_cos_add_sin`). The supporting calculus (`GradientOneForm`) includes the coordinate Cartan formula for exact forms (`lieOneForm_gradField`: the Lie derivative of dV along u is exact with potential $\langle u, \nabla V \rangle$, i.e. $L_u dV = d\langle u, \nabla V \rangle$ pointwise) and the self-transport identity (`lieOneForm_self`: $L_u u^\flat = (u \cdot \nabla)u + \frac{1}{2}\nabla|u|^2$). These two identities are what turn transport equations into Navier–Stokes with a pressure, next.

4.3 The pushdown theorem: Navier–Stokes with an explicit pressure

The structure `TransportedMomentumData` packages a velocity u , a gauge W , their certified time derivatives, the gradient/Laplacian exchange facts, and one transport hypothesis — the Itô-averaged frame-transport equation for the momentum one-form:

$$\partial_t u + \nabla \dot{W} + L_u(u^\flat + dW) = \nu \Sigma_{\text{frame}}(u^\flat + dW),$$

where Σ_{frame} is the normalized frame second-order sum of Section 4.2 (the hypothesis is stated in averaged form; no stochastic calculus is formalized, and the field is labeled accordingly in the source). From this single hypothesis the pushdown theorem derives the Navier–Stokes equations with a fully explicit pressure:

```

/-- Pushdown theorem: the averaged frame-transport equation for the
momentum one-form u^flat + dW forces u to satisfy the Navier-Stokes
equations with the explicit pressure
p = dW/dt + (1/2)<u,u> + <u, grad W> - nu * Lap W. -/
theorem navierStokes_of_transportMomentum (t : Real) (x : R3) :
  deriv (fun s => D.u s x) t + fderiv Real (D.u t) x (D.u t x) +
  gradField (D.pressure t) x =
  D.nu smul coordLaplacian (D.u t) x

```

with

$$p = \partial_t W + \frac{1}{2}\langle u, u \rangle + \langle u, \nabla W \rangle - \nu \Delta W.$$

The pressure is not posited: it is assembled from the exact terms of the transport (gauge drift, kinetic energy via the self-transport identity, gauge advection via the Cartan formula, viscous gauge

term), and the theorem is unconditional given the data. This is the repaired route’s replacement for v7’s identity-energy pushdown: the same endpoint (a genuine Navier–Stokes solution with pressure), reached without the false adjoint bound.

4.4 The vorticity-stretching gate and the conditional route

Continuation is fed by the averaged Cauchy formula $\omega(t) = \mathbb{E}[(DX_t)(\omega_0 \circ A_t)]$, which bounds $\|\omega(t)\|_\infty$ by moments of the flow Jacobian. The gate is the named open input of the coarse route, exactly:

```
-- The vorticity-stretching gate (named analytic input; OPEN). -/
def StochasticStretchingEstimate
  (D : StochasticCauchyVorticityData Omega) (T : Real) : Prop :=
  IntervalIntegrable D.expectedJacobianBound volume 0 T
```

Its consumers are mechanical and checked: a time-uniform Jacobian bound gives a time-uniform vorticity bound (`vorticity_uniform_bound_of_jacobian_bound`), and under the gate the BKM integrand $t \mapsto \|\omega(t, x)\|$ is interval-integrable with $\int_0^T \|\omega\| \leq \|\omega_0\|_\infty \int_0^T \mathbb{E}\|DX_t\|_\infty$ (`bkm_integrand_intervalIntegrable`, `bkm_integral_bound`). The assembled conditional route (`stochasticLagrangian_conditional_route`) states: given transported momentum data, averaged-Cauchy vorticity data linked by $\omega = \text{curl} u$, and the gate on $[0, T]$, the velocity satisfies Navier–Stokes with the explicit pressure *and* the BKM integrand is controlled. Every step is machine-checked; the only analytic input is the gate.

The route is inhabited. Non-vacuity is itself checked, twice. The rest state instantiates the route end-to-end (`conditional_route_rest`, with the gate discharged by a constant envelope). Beyond the rest state, the module `DecayingShearInstance` provides a nonzero exact solution of the full transported-momentum system: the decaying shear $u(t, y) = e^{-t} \sin(y_1) e_0$ with gauge amplitude $b(t) = (e^{-2t} - e^{-4t})/8$ on the frequency-doubled mode, whose balance $b' = -4b + \frac{1}{4}e^{-2t}$ (`gaugeAmp_balance`) absorbs the kinetic self-transport term; `decayingShear_navierStokes` confirms the pair satisfies Navier–Stokes with the representation pressure at $\nu = 1$. So the conditional machinery is not vacuously true: its hypotheses have nonzero exact witnesses.

4.5 The frozen-strain countermodel and the aperture geometry

Any proof of the gate must use more than local strain geometry plus stochastic averaging. The frozen-strain model makes this a theorem. For the prescribed strain $A = \text{diag}(2\gamma, -\gamma, -\gamma)$ and the pancake packet (vorticity along the expanding axis, wave vector along a compressive axis), the module `FrozenStrainModel` proves that

$$\omega(t, x) = a(t) \sin(k(t) x_1) e_0, \quad k(t) = k_0 e^{\gamma t}, \quad a(t) = a_0 \exp\left(2\gamma t - \nu k_0^2 \frac{e^{2\gamma t} - 1}{2\gamma}\right)$$

is an *exact* solution of the frozen vorticity equation $\partial_t \omega + (Ax \cdot \nabla)\omega = A\omega + \nu \Delta \omega$ (`frozenStrainVorticity_solves`; the drift, stretching, phase and viscous terms are all checked, not modeled). Writing $g = 2\gamma/(\nu k_0^2)$ for the normalized strain, the amplification peaks at $t^* = \log g/(2\gamma)$ with the exact value

$$\frac{a(t^*)}{a_0} = g e^{1/g-1} \quad (\text{frozenStrainAmplification_peak}),$$

which is unbounded over the strain family: no uniform depletion bound of the form “pancake alignment always acquires diffusive decay at rate $\nu\lambda^2$ ” survives

(frozenStrain_no_uniform_amplification_bound). Any correct gate proof must therefore use the *self-consistency* of the Navier–Stokes strain — a statement quantified over arbitrary imposed strains is false.

The angular version is the aperture-coverage geometry. Anisotropic Bernstein estimates control cone apertures up to $\lambda^{-1/2}$; stretch–diffusion covariance controls apertures above $\sqrt{\gamma/\nu}/\lambda$. The band claimed by neither,

$$(\lambda^{-1/2}, \sqrt{\gamma/\nu}/\lambda), \quad \text{is nonempty} \iff \nu\lambda < \gamma \quad (\text{anisotropic_coverage_gap}).$$

4.6 July additions: the aperture dichotomy is the strain budget

The July campaign sharpened the coverage gap into an equivalence and closed the remaining escape routes. In `ApertureCoverageDichotomy.lean`:

```
theorem pancakeApertureCovered_iff_strain_le
  (hnu : 0 < nu) (hlam : 0 < lam) :
  (forall theta : Real,
    theta <= (Real.sqrt lam)^{-1} \wedge Real.sqrt (gamma / nu) / lam <= theta)
  <-> gamma <= nu * lam
```

That is: the “narrow $\Rightarrow$ enstrophy-controlled, else $\Rightarrow$ angularly coercive” dichotomy tiles the whole aperture range *exactly when* the scale-local strain budget $\gamma \leq \nu\lambda_q$ holds. On the gap band both mechanisms fail quantitatively: `gapAperture_diffusionRate_lt_strain` proves $\nu(\theta\lambda)^2 < \gamma$ below the stretch–diffusion threshold (the diffusive rate is strictly beaten by the strain), and `gapAperture_bernstein_volume_exceeds_planar_budget` proves $\lambda^2 < \theta^2\lambda^3$ above the Bernstein threshold (the cone’s Fourier-support volume exceeds the planar budget behind the improved $L^2 \rightarrow L^\infty$ constant).

The time-averaged escape route is closed by a Grönwall-saturation computation: over the coherent window the packet consumes an integrated strain budget of exactly $\int_0^{t^*} 2\gamma dt = \log g$ (`frozenStrain_integratedStrain_peakWindow`) while amplifying to $g e^{1/g-1} \geq e^{\log g-1}$, i.e. amplification at least $e^{\text{budget}-1}$ (`frozenStrainAmplification_peak_ge_exp_integratedStrain_sub_one`). An averaged strain hypothesis therefore bounds amplification only exponentially in itself and cannot substitute for the pointwise-in-time aperture condition in a dyadic relaxation estimate.

4.7 The dyadic pancake closure: verified pieces and the isolated pin

The July repair proposal claims a complete closure of the gate: decompose the weighted stretching sum by angular sector, prove the pancake-sector estimate

$$\int_0^T \sum_{q \geq Q} \lambda_q^{-2} \|A_q \omega_q^{\text{pan}}\|_\infty dt \leq C(T, \nu, u_0, Q) + \varepsilon \int_0^T D_Q(t) dt, \quad D_Q(t) := \sum_{q \geq Q} \|\Delta_q \omega(t)\|_\infty,$$

with $A_q = S(S_{q-L}u)$ the low-frequency strain and ω_q^{pan} the projection onto the pancake cone of aperture $\delta_q = \lambda_q^{-1/2}$, then absorb $\varepsilon \int D_Q$ and conclude $\int_0^T D_Q < \infty$, hence BKM. The formalization adjudicates this claim piece by piece.

The genuinely new ingredient is verified. The proposal’s Step 1 — a pancake vorticity mode does not generate the strain that stretches itself — is checked exactly, not at symbol level. In `PlaneWaveSelfStrain`: for $\omega(x) = \alpha \sin(\kappa x_1) e_0$ with its Biot–Savart velocity (the shear $u(x) = -(\alpha/\kappa) \cos(\kappa x_1) e_2$), `curlField_planeWaveVelocityMode` proves $\text{curl } u = \omega$ exactly, and

`strainRateApply_planeWaveVelocityMode_vorticity` proves $S(u)\omega = \frac{1}{2}(\nabla u + (\nabla u)^\top)\omega = 0$ pointwise, with a non-vacuity witness (`planeWaveVorticityMode_ne_zero`). The scope is honest: the identity is for one exact mode; a pancake *sector* aggregates a cone of modes, and charging the cross-mode strain is precisely the open budget below.

The absorption skeleton is checked with explicit constants. The structure `DyadicVorticityStretchingData` carries the four time functionals (D_Q ; pancake, angularly coercive, and residual stretching sums), their nonnegativity and integrability, and the transported dyadic relaxation inequality. The three sector estimates are explicit predicates, e.g.

```
def PancakeSectorEstimate (C eps : Real) : Prop :=
  (Integral of D.pancakeStretchSum over 0..D.T) <=
    C + eps * (Integral of D.highBlockSum over 0..D.T)
```

(coercive and residual analogous; both are *interfaces* here — established in the source analysis by good-unknown/stretch-diffusion and Coifman–Meyer arguments respectively, consumed as named hypotheses, not re-proved). The absorption theorem `highBlockIntegral_le_of_sectorEstimates` proves: sector estimates with total gain $\varepsilon_p + \varepsilon_c + \varepsilon_r < 1$ imply

$$\int_0^T D_Q(t) dt \leq \frac{C_{\text{relax}} + C_p + C_c + C_r}{1 - (\varepsilon_p + \varepsilon_c + \varepsilon_r)},$$

with the a-priori finiteness carried by the integrability fields (supplied in the analytic setting by local-in-time regularity and continuation, which is standard and not re-derived). The theorem `pancakeSectorEstimate_of_scaleLocalStrainBudget` derives the pancake-sector estimate from the pointwise scale-local budget by time integration.

Where the claimed proof fails. The proposal’s high-low split charges the whole non-pancake complement to the angularly coercive estimate. By Section 4.6 that charge is *equivalent* to the scale-local budget $\gamma \leq \nu\lambda_q$ — which is exactly the bound the proposal’s own conditional section had identified as “the remaining burden” before the claimed proof. The claimed proof therefore consumes, at its high-low split, the statement it set out to prove; the closure is circular at precisely that point, and nothing else in it repairs the circle.

The remaining pin is a precise predicate, structurally verbatim:

```
-- The scale-local pancake strain budget (named analytic input; OPEN --
the first load-bearing missing lemma of the claimed pancake closure). -/
def ScaleLocalPancakeStrainBudget (D : DyadicVorticityStretchingData)
  (B eps : Real) : Prop :=
  exists budget : Real -> Real,
    IntervalIntegrable budget volume 0 D.T /\
    (Integral of budget over 0..D.T) <= B /\
    forall t in Set.Icc 0 D.T,
      D.pancakeStretchSum t <= budget t + eps * D.highBlockSum t
```

Here `pancakeStretchSum` represents $\sum_{q \geq Q} \lambda_q^{-2} \|A_q \omega_q^{\text{pan}}(t)\|_\infty$. The obligation is to establish this pointwise envelope for the *self-consistent* Navier–Stokes strain (generated from the vorticity by the Biot–Savart law), not for an externally imposed frozen strain — via a genuinely Navier–Stokes-specific mechanism (pressure/Hessian feedback, multi-scale Biot–Savart self-depletion, or frequency-cascade accounting, in the proposal’s own taxonomy). The verified plane-wave null structure is evidence in the right direction; the aperture equivalence states exactly what quantitative multi-scale estimate is still missing.

The sharpened conditional route. The headline theorem `stochasticLagrangian_conditional_route_of_pancakeBudget` closes the loop: given transported momentum data, dyadic relaxation data whose block sum dominates the high-frequency part of $\|\text{curl } u\|$ (the Littlewood–Paley decomposition interface), the scale-local pancake strain budget, and the coercive/residual interfaces with total gain below 1, the velocity satisfies Navier–Stokes with the explicit pressure and, for each fixed spatial point, the vorticity norm is interval-integrable with an x -independent bound. This reaches the same pointwise endpoint as the coarse conditional route with the gate replaced by the sharper geometric pin; the two pins are cross-referenced in the sources and both remain open. The common time majorant is the correct precursor to the BKM criterion, but the current theorem does not yet define the spatial essential supremum, prove its measurability and integrability, or invoke a formal continuation theorem. An end-to-end rest-state canary (`rest_conditional_route`) checks that the refined hypotheses are simultaneously satisfiable.

5 The direction-coherence refinement of the pin

The September 2026 campaign adjudicated the Biot–Savart self-depletion mechanism of the pancake budget by pre-registered kinematic experiment, then distilled and proved the surviving content, and committed a strictly sharper pin. All statements below are machine-checked with the standard three axioms; the labs are `ns_pancake_kinematic_depletion_lab.py` and `ns_pancake_coherent_aggregate_lab.py` in the validation lab, with pre-registered verdict criteria recorded in the files before execution.

The dichotomy. Wavevector-cone confinement alone does not deplete pairwise stretching. In the exact lattice interaction model (the amplitude-level Fourier symbol of $S(u)\omega$ with Biot–Savart velocity), the aggregate cone gain equals the cone measure (measured $D \sim \delta^{2.06}$ across dyadic shells), while the worst pair saturates the generic maximum at every aperture. The distilled exact statement (`PancakeConeDirectionDichotomy`): the axial source $k_1 = (0, 0, m)$ with vorticity e_x stretches the *perpendicular* in-plane vorticity direction e_y of the cone receiver $k_2 = (1, 0, M)$ with $\sigma^2 = 1/4$ exactly, for all m, M ; the *aligned* polarization $(-M, 0, 1)$ of the same wavevector pair is depleted at exactly $\sigma^2 = 1/(4(1 + M^2))$. The operative small parameter of the self-depletion mechanism is therefore the vorticity-direction misalignment inside the pancake — the Constantin–Fefferman coherence quantity — not the wavevector cone.

The coherent pair estimate (proved). Give every mode the direction-coherent polarization: the fixed in-plane reference e_x projected off the wavevector. Then (`PancakeCoherentPairEstimate.sigmaSq_coherent_1e`) for wavevectors inside pancake cones of squared apertures $d_1, d_2 \leq 1/4$,

$$\sigma^2 \leq 6(d_1 + d_2),$$

with the mechanism carried by the exact identity $k \times (|k|^2 e_x - (k \cdot e_x)k) = |k|^2(0, k_z, -k_y)$: the Biot–Savart velocity of a coherent mode is perpendicular to the reference direction. The bound aggregates over any finite family of coherent sources (`sum_sigmaSq_coherent_1e`), matching the measured coherent-aggregate gain of the second lab.

Amplitude and field-transfer layer (proved, but not yet complete). The module `PancakeCoherentWeightedTransfer` inserts the actual scalar Fourier amplitudes into the pair estimate. It proves the cardinality-free finite convolution bound

$$\left\| \sum_{k \in E} T_k(a_k p_k, b p_\ell) \right\|_{\ell^1(\mathbb{C}^3)} \leq 6(\delta_1 + \delta_2) \left(\sum_{k \in E} |a_k| \right) |b|,$$

and its Cauchy–Schwarz energy form, where $p_k = P_k e_0$. The exact theorem `PancakeConeLatticeCount.card_le_pancakeBox` shows that, at $\lambda = N^2$ and aperture $1/N$, the cone lies in a box of cardinality $(4N+1)^2(4N^2+1) = O(\lambda^2)$, rather than the ambient $O(\lambda^3)$ count.

The modules `PancakePeriodicCoherentSplit` and `PancakePeriodicComplexStretch` then move from the rational symbol model to the genuine Fourier lattice $\mathbb{Z}^3$ and arbitrary complex coefficients. For every cone mode they construct the exact orthogonal split

$$\hat{\omega}(k) = \frac{\langle P_k e_0, \hat{\omega}(k) \rangle}{\langle P_k e_0, P_k e_0 \rangle} P_k e_0 + \hat{\omega}_{\text{mis}}(k), \quad \langle P_k e_0, \hat{\omega}_{\text{mis}}(k) \rangle = 0,$$

prove that both terms remain divergence-free, define the complexified Biot–Savart strain symbol, and prove its agreement with the rational symbol. The aperture estimate and the $O(\lambda^2)$ energy bound therefore hold for arbitrary complex coherent amplitudes, not only rational test vectors.

The module `PancakeRealCoherentPairEstimate` removes the rational-grid restriction needed for a smooth multiplier. It proves the same cancellation for all real frequencies, constructs the rank-one orthogonal coherent projection of an arbitrary amplitude, proves that this projection does not increase Euclidean energy, and obtains the field-facing pointwise estimate against the energies of the original input amplitudes.

The first attempted receiver-only endpoint has now been fenced off rather than promoted. The generic modules `PancakeConeKernelScaling` and `PancakeConeOperatorKernel` correctly prove that $K_\delta(x_\perp, z) = \delta^3 K(\delta x_\perp, z)$ has $\|K_\delta\|_{L^1} = \delta \|K\|_{L^1}$, but one of those three powers must already be a genuine symbol gain; only two come from the transverse Jacobian. The periodic theorem `PancakePeriodicOperatorKernel` now proves the corresponding no-go: if an exact projection fixes a unit Fourier mode, its kernel has L^1 mass at least one. Thus the smooth family in `PancakeSmoothCoherentAperture`, whose symbol was explicitly attenuated by an additional factor δ , is a useful analytic model but is not the desired coherent field projection.

The genuine aperture factor has instead been located in the complete bilinear interaction. For $k_1 = (\delta a, \delta b, c)$ and $k_2 = (\delta d, \delta e, f)$, `PancakeCoherentStretchFactorization` proves exactly, both for the canonical coherent vectors and for coherent projections of arbitrary input amplitudes,

$$T(k_1, k_2) = \delta T_{\text{norm}}(\delta, a, b, c, d, e, f),$$

where T_{norm} is smooth through $\delta = 0$ before the projection denominators are introduced. The module `PancakeSmoothCoherentStretchSymbol` supplies nested cutoffs keeping both axial frequencies away from zero, constructs all 27 entries of the localized normalized tensor as compactly supported Schwartz symbols on six frequency variables, proves the tensor is nonzero, and gives an integrable inverse-Fourier kernel with exact entrywise and tensor Fourier recovery for each fixed δ . Finally, `PancakeBilinearKernelScaling` proves the correct four-transverse variable scaling

$$K_\delta(y_\perp, z_\perp; y_\parallel, z_\parallel) = \delta^5 K(\delta y_\perp, \delta z_\perp; y_\parallel, z_\parallel),$$

whose four-dimensional Jacobian leaves one power of δ , together with the resulting bilinear $L^\infty \times L^\infty \rightarrow L^\infty$ Young bound. The module `PancakeUniformKernelFamily` differentiates the symbol jointly in δ and all six frequency variables, uses its fixed compact support to obtain uniform Schwartz seminorms, converts those bounds into uniform polynomial decay of every inverse-Fourier entry, and sums the 27 entries to prove

$$\sup_{0 \leq \delta \leq 1/2} \int_{\mathbb{R}^6} \|K_{\text{norm}, \delta}(y, z)\| dy dz < \infty.$$

Thus the normalized Euclidean kernel-mass obligation is discharged. The module **PancakeBilinearPeriodization** then carries this kernel to the standard six-torus. It proves the fundamental-domain unfolding lemma, matches the torus characters to the Euclidean Fourier normalization, and shows for every $n \in \mathbb{Z}^6$ that

$$\widehat{K_{\text{norm},\delta}^{\text{per}}}(n) = M_{\text{norm},\delta}(n).$$

It also proves that the resulting six-dimensional kernel acts on a pair of three-torus Fourier monomials through this exact coefficient. Restoring the algebraically extracted factor gives a periodic kernel realizing the localized coherent-stretching symbol and one constant C , independent of $0 \leq \delta \leq 1/2$, such that

$$\|B_\delta(f, g)\|_{L^\infty(\mathbb{T}^3)} \leq 9\delta C \|f\|_{L^\infty(\mathbb{T}^3)} \|g\|_{L^\infty(\mathbb{T}^3)}.$$

Thus Euclidean-to-torus transference and the frequency-uniform frozen-axis coherent aperture bound are discharged. The module **PancakeFiniteFourierBilinearTransfer** distributes the bilinear action over arbitrary finite trigonometric polynomials and proves the exact double Fourier sum without a mode-cardinality loss. It also identifies the assembled 27-entry symbol, on the retained chart, with the complex-bilinear extension of the coherently projected normalized Biot–Savart stretching interaction; off that chart both sides vanish. Thus the periodic kernel now acts exactly as the localized coherent PDE interaction on finite Fourier fields.

The module **PancakeDyadicKernelScaling** carries this base chart to actual square-dyadic shells. For every positive r it proves

$$\int_{\mathbb{R}^6} \|r^6 K(rx)\| dx = \int_{\mathbb{R}^6} \|K(x)\| dx, \quad r^6 \widehat{K(r \cdot)}(\xi) = \widehat{K}(\xi/r),$$

and proves the corresponding exact fundamental-cell coefficient after periodization. It then constructs a volume-preserving regrouping of the six variables into four transverse and two axial coordinates and proves

$$\int_{\mathbb{R}^6} \|\delta^5 K(\delta x_\perp, x_{\text{ax}})\| dx = \delta \int_{\mathbb{R}^6} \|K(x)\| dx, \quad \widehat{K}_\delta(\xi) = \delta \widehat{K}(\xi_\perp/\delta, \xi_{\text{ax}}).$$

At aperture $\delta = 1/N$ and shell scale $r = N^2$, the reciprocal argument is machine-checked to be precisely the source/receiver chart $(k_1/N, k_2/N, k_3/N^2)$ (and similarly for the second input). Consequently the periodized square-dyadic kernel has both the exact coefficient

$$\widehat{K}_N^{\text{per}}(k, l) = \frac{1}{N} M_{\text{norm}, 1/N} \left(\frac{k_1}{N}, \frac{k_2}{N}, \frac{k_3}{N^2}, \frac{l_1}{N}, \frac{l_2}{N}, \frac{l_3}{N^2} \right)$$

and the uniform fundamental-cell estimate $\int \|K_N^{\text{per}}\| \leq C/N$ for every $N \geq 2$. Thus the frozen-axis anisotropic square-dyadic transfer is discharged. The full-annulus and variable-frame assembly is described below.

The module **PancakeSquareDyadicFiniteTransfer** descends this final kernel through the canonical half-open representative of the unit six-torus. It proves that torus Fourier coefficients and torus L^1 mass are exactly the corresponding fundamental-cell quantities, and hence that arbitrary finite Fourier fields are acted on by the exact double sum of square-dyadic coherent interactions. The endpoint estimate is cardinality-free:

$$\|B_N(f, g)\|_{L^\infty(\mathbb{T}^3)} \leq 9 \frac{C}{N} \|f\|_{L^\infty(\mathbb{T}^3)} \|g\|_{L^\infty(\mathbb{T}^3)}, \quad N \geq 2.$$

Thus the finite-field frozen-axis square-dyadic transfer is also discharged.

The module `PancakeFrozenFrameKernelTransfer` performs the frozen-frame step at the kernel level. Because a general rotation does not preserve $\mathbb{Z}^3$, it first rotates the Euclidean cover kernel and only then periodizes on the fixed standard lattice. Orthogonal change of variables preserves the exact cover L^1 mass, while Fourier covariance evaluates the symbol at the rotated reciprocal frequency. The module constructs a genuine Euclidean linear isometry from every algebraic `OrientedFrameEquiv`, lifts it diagonally to the two interacting frequencies, and proves the exact finite-Fourier identity with the retained coherent Biot–Savart interaction in the resulting oriented square-dyadic chart. One constant works for all such frames:

$$\|B_{N,F}(f, g)\|_{L^\infty(\mathbb{T}^3)} \leq 9 \frac{C}{N} \|f\|_{L^\infty(\mathbb{T}^3)} \|g\|_{L^\infty(\mathbb{T}^3)}, \quad N \geq 2.$$

Thus frozen oriented-frame transference of the exact square-dyadic kernel is discharged. Physical vector conjugation, pointwise frame variation, and the replacement of the initial demonstration chart by the full annulus are handled next.

The module `PancakePhysicalFrameKernelTransfer` supplies the missing input/output conjugation. It complexifies the algebraic real frame, proves that complexification commutes with inversion, and bounds both the forward and inverse maps by the dimension-only coordinate constant 3. Applying the frequency-rotated kernel to the two frame-coordinate inputs and rotating the output back therefore gives

$$\|B_{N,F}^{\text{phys}}(f, g)\|_{L^\infty} \leq 243 \frac{C}{N} \|f\|_{L^\infty} \|g\|_{L^\infty}.$$

The exact coherent finite-Fourier formula is proved with these three frame maps in place. More importantly, the same estimate is proved for arbitrary continuous bounded fields and then for an arbitrary pointwise frame field $F = F(x)$, simply by instantiating the frame-uniform kernel estimate at the observation point. No derivative of F occurs. This removes the variable-frame commutator as an analytic loss at this particular pointwise L^∞ endpoint.

The modules `PancakeAnnularChartCutoff`, `PancakeAnnularUniformKernel`, and `PancakeAnnularMultiplierAssembly` remove the remaining demonstration-chart restriction. On a square-dyadic annulus the cone inequality forces each normalized axial coordinate into $[-2, -3/4] \cup [3/4, 2]$, while the transverse coordinates lie in $[-2, 2]$. There are therefore exactly four source/receiver sign combinations. Smooth compact buffers are constructed for them, their coherent denominators are proved globally nonzero, and their inner cutoffs satisfy the exact partition identity

$$\chi_{++} + \chi_{+-} + \chi_{-+} + \chi_{--} = 1$$

throughout the normalized annulus. The parametric Schwartz estimate is proved for every buffered cutoff and yields one aperture-uniform inverse Fourier L^1 bound for the assembled four-chart kernel. Its Fourier symbol is machine-checked to equal the *uncut* coherent Biot–Savart stretching interaction on the whole annulus.

Finally, `PancakeAnnularFrozenFrameTransfer` and `PancakeAnnularPhysicalFrameTransfer` carry this assembled multiplier through the rotated Euclidean cover, square-dyadic rescaling, periodization, physical input/output conjugation, and arbitrary pointwise frame fields. The exact finite-Fourier identity and the cardinality-free estimate

$$\|B_{N,F(x)}^{\text{ann}}(f, g)(x)\| \leq 243 \frac{C}{N} \|f\|_{L^\infty} \|g\|_{L^\infty}$$

hold uniformly for $N \geq 2$. Thus finite sign-chart coverage and partition assembly are discharged. The module `PancakeAnnularSectorProjector` then defines the concrete finite Fourier

projector obtained by filtering a given support with the normalized pancake-annulus predicate. It proves, coordinate by coordinate, that the six-dimensional frozen square-dyadic chart of a selected pair is exactly the pair of their independently selected three-dimensional normalized modes. Consequently the full-annulus premise of the exact physical kernel formula is automatic on these projected fields; it is no longer a free support hypothesis. The subsequent module **PancakeSquareDyadicSectorProjector** narrows this enclosing selector to the genuine microlocal sector: radial frequency in $[N^2, 2N^2)$ and squared transverse aperture at most N^{-2} after rotation into the chosen frame. From these physical inequalities it proves the transverse bounds and the nontrivial axial lower bound $3/4$, hence inclusion in the four-chart annulus. The exact coherent-action formula therefore applies directly to the concrete finite shell-and-cone projections. The modules **PancakeAxialFrameConstruction** and **PancakeStrainAxialFrame** then remove the caller-supplied pointwise frame. The first constructs, by two explicit reflections, an orientation-preserving orthogonal map sending any unit direction to the axial vector; the second feeds it the smallest-eigenvalue direction supplied by the ordered spectral theorem for the symmetric strain. Thus the pointwise C/N endpoint now applies directly to a frame constructed from the strain. Measurable dependence of the spectral choice through eigenvalue collisions, and identification of these concrete finite projectors with the actual Littlewood–Paley coherent share, are still required.

The modules **PancakeComplexMisalignmentExpansion**, **PancakeComplexCoherentPDEIdentification**, and **PancakePhysicalMisalignmentSplit** close the algebraic split behind that remaining analytic task. The former expands the actual complex Biot–Savart stretching interaction into the coherent–coherent term and the three terms containing at least one misalignment factor, first modewise and then under arbitrary finite source/receiver Fourier sums. Thus the finite-field norm analogue of **split_1e** follows from an exact identity and one triangle inequality. The latter proves that N^{-1} times the normalized annular symbol is literally the complexified PDE stretching operator applied to the two coherent chart projections. On the genuine rotated square-dyadic sector, the exact physical kernel formula is therefore rewritten directly as a sum of coherent–coherent PDE interactions; no analogy between two separately defined expressions remains. The third module performs the sum-level substitution: for arbitrary finite Fourier data it proves the exact physical identity

$$\mathcal{S}_{N,F} = \mathcal{K}_{N,F}^{\text{coh}} + \mathcal{R}_{N,F}^{\text{mis}},$$

where the remainder is precisely the sum of the coherent–misaligned, misaligned–coherent, and misaligned–misaligned interactions. A triangle inequality and the annular kernel theorem give

$$|\mathcal{S}_{N,F}(x)| \leq 243(C/N)MP + |\mathcal{R}_{N,F}^{\text{mis}}(x)|,$$

with no dependence on the two Fourier support cardinalities. The same theorem is specialized to the pointwise frame built from the most compressive strain eigenvector, so this statement no longer accepts a caller-supplied frame. The follow-up module **PancakeFullStrainFrame** fixes the residual rotation around the compressive axis by an explicit orientation-preserving planar rotation. It proves that the inverse complex image of e_x is exactly the most expanding strain eigenvector and repeats the exact split and C/N bound in this full eigenline-aligned frame. Consequently the Fourier remainder’s coherent reference is now the same expanding line used by the invariant alignment-energy dynamics below.

The module **PancakeMisalignmentEnergyBridge** makes that compatibility quantitative. If a complex Fourier amplitude u is divergence free at frequency k , its exact coherent-line remainder $u - P_k u$ satisfies

$$|u - P_k u|^2 \leq |u_y|^2 + |u_z|^2$$

in the full strain chart, where the chart's x -axis is the expanding eigenline. The proof applies the real orthogonal-projection inequality to the real and imaginary parts and proves that complexification of the oriented frame preserves the divergence pairing. Summing over the actual integer modes selected by the physical square-dyadic projector introduces no cardinality factor. An exact finite Parseval theorem then gives

$$\sum_{k \in \Gamma_{N,F}} |u_k - P_k u_k|^2 \leq \int_{\mathbb{T}^3} (|U_y(x)|^2 + |U_z(x)|^2) dx,$$

where U is the finite reconstruction in one frozen full strain frame. This closes the coefficient-to- L^2 bridge; it does not by itself control a frame varying with x , nor does an L^2 estimate alone supply the L^∞ -based BKM integrand.

The follow-up module **PancakeTransverseEnergyFreezing** supplies the local varying-frame comparison without choosing eigenvector signs. For symmetric strains A, B with $\|A - B\| \leq \varepsilon$, it first proves the top-eigenvalue Weyl bound

$$|\lambda_0(A) - \lambda_0(B)| \leq \|A - B\|.$$

It then applies the matched-eigenline projector estimate. For every positive threshold, either a frozen top gap of B is smaller than threshold $+$ ε , or

$$\sum_{k \in \Gamma_{N,A}} |u_k - P_{A,k} u_k|^2 \leq \int_{\mathbb{T}^3} |U_{\perp B}(x)|^2 dx + 4 \frac{\varepsilon}{\text{threshold}} \int_{\mathbb{T}^3} |U(x)|^2 dx.$$

Thus the eigenvector-sign and eigenvalue-collision problems are no longer hidden in a continuity premise. The remaining task is to aggregate this cellwise alternative with the same-strain small-gap coercivity and the space-time Littlewood–Paley budget.

The module **PancakeTransverseEnergyCoercivity** closes the separated cell algebra. If S is symmetric, e_0 is its top eigenvector, and both top gaps dominate g , then for every vector u —including $u = 0$ —

$$g(|u|^2 - |u \cdot e_0|^2) \leq \lambda_0(S)|u|^2 - u \cdot Su.$$

The proof is the three-dimensional spectral expansion and is deliberately homogeneous, so it never divides by vorticity magnitude. Applying it to the real and imaginary parts, integrating the finite reconstruction, and composing with the freezing theorem yields the checked cellwise dichotomy: either a frozen top gap is below threshold $+$ ε , or the gap-weighted exact Fourier remainder is bounded by the integrated complex Rayleigh defect plus $(\text{threshold} + \varepsilon)4\varepsilon/\text{threshold}$ times the full sector energy. The remaining issue is no longer the separated-cell estimate itself, but its bounded-overlap space-time aggregation and endpoint scaling.

The finite combinatorics of that aggregation is now checked in **PancakeCellAggregation**. It packages one frozen comparison datum, filters an arbitrary finite family into collision and separated cells, and sums the separated coercive estimates with no support-cardinality constant. Nonnegativity of both the complex Rayleigh defect and full energy allows the right-hand sums to be enlarged to the whole family. Two generic analytic lemmas prove the corresponding bounded-overlap transfer: for measurable hard cells of multiplicity at most M ,

$$\sum_i \int_{Q_i} f \leq M \int f,$$

and for nonnegative smooth weights χ_i satisfying $\sum_i \chi_i \leq M$, the same conclusion holds for varying local densities dominated by one global density. This does *not* yet identify the global Parseval energy

in each frozen comparison with a hard- or smooth-localized cell energy. A spatial cutoff changes Fourier support, so that step requires a genuine localization/commutator or almost-orthogonality estimate; bounded overlap alone cannot supply it.

The first part of that localization junction is now exact rather than an interface. The module `PancakeFiniteFourierLocalization` proves that for finite Fourier cutoff χ and vector field U ,

$$\chi(x)U(x) = \sum_{p,k} e^{2\pi i(p+k)\cdot x} \hat{\chi}(p) \hat{U}(k).$$

Thus the localized output support is precisely contained in the Minkowski sum of the cutoff and field supports, and every output mode differs from an original sector mode by an actual cutoff frequency. It also proves directly in physical space that $\|\chi U\|_{L^2}^2 \leq C^2 \|U\|_{L^2}^2$ whenever $\|\chi\|_{L^\infty} \leq C$; there is no mode-count loss. Additivity of the normalized oriented-frame chart then proves that cutoff support bounded by transverse and axial margins $(\rho_\perp, \rho_\parallel)$ moves the original annulus into the explicit buffered box

$$|v_{1,2}| \leq 2 + \rho_\perp, \quad \frac{3}{4} - \rho_\parallel \leq |v_3| \leq 2 + \rho_\parallel.$$

For the fixed margins $(\rho_\perp, \rho_\parallel) = (1/4, 1/8)$, the module `PancakeBufferedAnnularMultiplier` now completes the next part of this junction. It enlarges the signed retained axial gates from radius $3/4$ to $7/8$, while leaving their support radius 1 and the old denominator-safe outer gates unchanged. The resulting four cutoffs form an exact partition on the whole leaked box

$$|v_{1,2}| \leq \frac{9}{4}, \quad \frac{5}{8} \leq |v_3| \leq \frac{17}{8}.$$

The assembled tensor symbol therefore equals the uncut coherent stretching symbol on every permitted localized output pair. Applying the generic buffered-cutoff theorem chartwise and summing gives one aperture-uniform L^1 bound for the assembled inverse kernel.

The modules `PancakeBufferedAnnularFrozenFrameTransfer` and `PancakeBufferedAnnularPhysicalFrameTransfer` carry this enlarged operator through anisotropic aperture scaling, square-dyadic rescaling, frozen rotation, periodization, and physical input/output conjugation. They prove the exact torus coefficient formula and retain the cardinality-free C/N endpoint, uniformly even when a different frozen frame is selected at each output point. Finally, `PancakeLocalizedBufferedTransfer` collects the repeated cutoff/field pairs by their output frequency and proves the exact physical coherent identity directly on products χU and ψV . Its premises mention only the original annular modes and the finite-band cutoff margin; the leaked-support condition is derived rather than assumed. It also proves the expected endpoint with losses only $\|\chi\|_\infty \|U\|_\infty$ and $\|\psi\|_\infty \|V\|_\infty$, never a support cardinality.

This closes the finite-band localization junction. A nonconstant smooth spatial cutoff still has an infinite Fourier tail, so the operator side of its finite approximation is checked in `PancakeBilinearActionStability` and `PancakeBufferedLocalizationStability`: for the physical buffered operator $\mathcal{B}_N$,

$$\|\mathcal{B}_N(f, g) - \mathcal{B}_N(f_0, g_0)\| \leq \frac{243C}{N} (\|f - f_0\|_\infty \|g\|_\infty + \|f_0\|_\infty \|g - g_0\|_\infty).$$

A further specialization transfers uniform scalar cutoff-approximation errors for $\chi - \chi_0$ and $\psi - \psi_0$ directly into this bound. Thus only the cutoff-specific Fourier decay input remained after that module. More precisely, `PancakeSmoothCutoffFourierTail` proves that every continuous torus cutoff with absolutely summable Fourier coefficients has, for each $\varepsilon > 0$, a finite Fourier

approximation within ε uniformly in space. It further proves that one such finite support lies in the fixed normalized cutoff-margin box at all sufficiently large square-dyadic scales. The scale threshold is uniform over every oriented frame because rotations preserve the Euclidean norm.

That remaining cutoff construction is now discharged in **PancakeConcreteSmoothCellCutoff**. A fixed compactly supported smooth real bump is periodized on the unit circle and tensored over three coordinates. Poisson transference identifies its Fourier coefficients with integer samples of a Schwartz Fourier transform, proving absolute summability. Translating this cutoff over an explicit $5^3 = 125$ quarter-grid produces a finite smooth cover: at every torus point at least one cutoff equals one, the summed quadratic weights have explicit uniform lower and upper bounds, and every translated cutoff retains absolute Fourier summability. One common scale therefore supplies simultaneous finite-Fourier approximations of all 125 cells. The module **PancakeConcreteSpatialCoverTransfer** feeds those approximations through the buffered stability theorem. For every pair of actual smooth cells and every sufficiently large scale, the localized physical interaction differs from the named coherent finite convolution by the explicit $O(\varepsilon/N)$ error, uniformly in the cells, frame, and support sizes. At this junction only the passage from the hard annular selector to the route's chosen smooth Littlewood–Paley projector remains.

The new module **PancakeArbitraryDataBlocks** removes the separate finite-ambient-support restriction. It enumerates complete hard radial shells directly in the integer lattice, proves that every nonzero mode belongs to a unique ordinary dyadic shell, and shows that each ordinary shell is covered without a gap by two adjacent square-dyadic multiplier shells. The lower and upper pieces split exactly into pancake and nonpancake modes, both as Fourier polynomials and as Parseval energies. Applied to an arbitrary periodic field, the pancake pieces satisfy the exact coherent multiplier formula and the actual smooth-cell $O(\varepsilon/N)$ transfer above; the ambient field need not have finite Fourier support. For a jointly continuous time-dependent field, every shell-energy profile is continuous and interval-integrable. At the natural L^2 level, vector Parseval and an explicit reindexing prove

$$\|u\|_{L^2}^2 = |\hat{u}(0)|^2 + \sum_{q \geq 0} \|\Delta_q u\|_{L^2}^2,$$

with summability of the nonnegative shell series; for mean-zero data the first term vanishes. This is the complete hard Littlewood–Paley energy ledger, not a finite-support surrogate.

The companion module **PancakeArbitraryDataMisalignment** connects these complete blocks to the exact PDE stretching split. On every complete square-dyadic pancake block, the full physical stretching sum is the coherent kernel action plus the exact three-term misalignment remainder, and the coherent term keeps its support-cardinality-free C/N bound. For divergence-free data, the squared misalignment coefficients are controlled by the physical transverse energy in the expanding strain frame; their frozen-frame time profile is continuous and interval-integrable whenever the Fourier coefficients are continuous. The collision-safe frozen-strain dichotomy therefore now applies to complete arbitrary-data blocks rather than an externally supplied finite support.

Finally, **PancakeConcreteCoverEnergy** performs the first actual cell-aggregation step. For every continuous nonnegative density f ,

$$\int f \leq \sum_{j=1}^{125} \int |\chi_j|^2 f \leq 125 \|\chi\|_\infty^2 \int f.$$

This is specialized to complete pancake-block energy, transverse misalignment energy, and frozen-strain Rayleigh defect. Spectral-gap coercivity is also proved after multiplication by each genuine smooth-cell weight, and summing the weighted inequalities introduces no cell-count factor beyond the explicit fixed cover comparison. The module **PancakeConcreteWeightedFreezing** then allows

the frozen strain B_j to vary with the cell. If every B_j is within ε of the observation strain and its top gap dominates threshold $+\varepsilon$, the complete arbitrary-data misalignment energy satisfies

$$(\text{threshold} + \varepsilon) M \leq \sum_{j=1}^{125} \int |\chi_j|^2 D_{B_j} + (\text{threshold} + \varepsilon) \frac{4\varepsilon}{\text{threshold}} 125 \|\chi\|_\infty^2 \int |u_{\text{pan}}|^2.$$

Thus varying the frozen strain from cell to cell introduces neither a frequency-support cardinality nor an additional cell-count loss. What remains is to construct these B_j from the true space-time strain with the required oscillation bound and to treat cells entering the small-gap branch.

The module **PancakeSpectralProjectorRegularity** addresses the measurability issue on the complementary separated branch. Writing $P_{\max}(S)$ for the rank-one projector onto the expanding line and $g(S) = \lambda_0(S) - \lambda_1(S)$, it proves that if $\|A - B\| < g(B)/2$, then

$$\|P_{\max}(A) - P_{\max}(B)\| \leq \frac{8}{g(B)} \|A - B\|.$$

Consequently $P_{\max}(S(x, t))$, and the complex transverse-energy density formed from it, are continuous and measurable on every region where a continuous strain field has positive top gap. This avoids both arbitrary eigenvector signs and any false continuity claim through a collision.

The module **PancakeSpectralClusterDichotomy** then resolves the topology of the top-gap collision branch. It proves the bottom-eigenvalue Weyl bound and local Lipschitz regularity of the projector onto the expanding two-plane (the complement of the bottom eigenline) wherever $g_{\text{bot}} = \lambda_1 - \lambda_2 > 0$. It also proves the exact defect identity

$$D_S(u) = g_{\text{top}} |u_1|^2 + (g_{\text{top}} + g_{\text{bot}}) |u_2|^2,$$

with the corresponding real/imaginary complex-coordinate energies. Finally, for trace-free strain, if both adjacent gaps are below τ , then $|\lambda_i| < 2\tau$ for all three eigenvalues and

$$\langle u_{\text{re}}, Su_{\text{re}} \rangle + \langle u_{\text{im}}, Su_{\text{im}} \rangle \leq 2\tau |u|^2.$$

Thus a small top gap now has two explicit subbranches: a stable expanding plane, or uniformly small stretching. The unresolved collision geometry is specifically the dynamical coherence problem inside that stable plane.

The module **PancakeExpandingPlaneResidence** supplies the first dynamic estimate on that branch. For a unit vorticity direction ξ and a unit bottom eigenvector e_2 , define the sign-invariant contracting-line energy $C = \langle \xi, e_2 \rangle^2$. This is exactly the energy escaping the stable expanding-plane projector. The checked spectral identity gives

$$g_{\text{bot}}(1 - C) \leq \langle \xi, S\xi \rangle - \lambda_2.$$

Consequently, while $C \leq 1/2$, the direction equation yields the integrated residence estimate

$$\int_0^T g_{\text{bot}}(t) C(t) dt \leq C(0) + 2 \int_0^T |\langle d, e_2 \rangle + \langle \xi, \dot{e}_2 \rangle| dt,$$

where d is the non-strain part of the direction rate. This proves that strain damps leakage into the contracting line even when the two expanding eigenvalues collide. It deliberately does not replace the two right-hand forcing channels by an a priori Navier-Stokes bound; doing so, and controlling coherence between directions inside the expanding plane, is the remaining dynamic content rather than a consequence of spectral algebra alone.

There is a second field-level issue which a fixed-axis multiplier does not address. The actual pancake frame is the eigensystem of the low-frequency strain and therefore depends on (x, t) . The module `PancakeFrameCovariance` proves that the continuous coherent estimate is rotation-covariant for every frozen oriented orthonormal frame, including arbitrary projected amplitudes. The module `PancakeVariableFrameCommutator` then gives the exact finite-Fourier identity

$$T_{m(x, \cdot)} f = T_{m(x_0, \cdot)} f + T_{m(x, \cdot) - m(x_0, \cdot)} f$$

and bounds the second term by the frequencywise symbol oscillation times the actual Fourier-amplitude mass. The uniform frozen-axis periodic kernel bound is now supplied by `PancakeBilinearPeriodization`; exact finite-field summation and the full anisotropic square-dyadic scaling are supplied by the two subsequent modules above; frozen oriented-frame kernel transference is supplied by `PancakeFrozenFrameKernelTransfer`; the full four-sign annulus and its physical variable-frame endpoint are supplied by the annular modules above, and the pointwise frame is now constructed from the strain. The hard dyadic time profiles, their pointwise Parseval identities, and their interval integrability for jointly continuous fields are now explicit. The remaining field-transfer pieces are aggregation of the proved frozen-cell/small-gap alternative over a physical space-time cover (the separated line and stable-plane projectors are now measurable, and the triple-collision branch has small stretching), passage to the intended smooth projectors, and identification of the hard pancake/nonpancake split with the route's dynamically defined coherent/misalignment functional. The older frequencywise commutator estimate remains correct, but the new uniform physical-kernel endpoint shows that no frame-oscillation estimate is needed merely to obtain the pointwise L^∞ bound. They are not silently included in the finite-sector estimate, and `coherent_le` has not yet been instantiated for the time-dependent Navier–Stokes data structure.

There is also a concrete scale warning at this junction. The present 5^3 smooth cover is a fixed-scale localization device. It gives a valid bounded-overlap energy comparison and uniform Fourier-cutoff transfer, but it cannot by itself force an arbitrarily oscillatory strain field to be ε -close to a constant on each cell. Instantiating the weighted freezing theorem therefore requires a scale-refined cover with overlap bounds independent of the refinement (and compatible cutoff-frequency margins), or a pointwise variable-strain argument. Treating the fixed 125 cells as if they automatically supplied arbitrary strain oscillation would be an invalid closure.

The refined pin (committed). The module `MisalignmentRefinedPin` splits the pancake-sector functional into a direction-coherent share, carrying the interface form of the proved aperture-gain kinematics, and a misalignment share; the new pin `MisalignmentStrainBudget` demands the time-integrable envelope only for the misalignment share, and the sealed reduction `scaleLocalPancakeStrainBudget_of_misalignmentBudget` reproduces the original budget with gain $\text{coherentGain} + \varepsilon$. The refinement is strict: the open burden moved from the full pancake stretching sum to its misalignment share, whose controlling quantity is the direction variation. The vorticity-direction equation exposes a dynamical handle on this quantity, but does not by itself give the required a priori estimate.

Exact unproven boundary. (i) Completion of the field-level transfer: the modewise complex split, amplitude estimate, lattice count, exact bilinear aperture factor, fixed- δ smooth tensor kernel, bilinear scaling/Young bound, exact finite torus realization, frozen-frame covariance, and variable/frozen split are now proved. Uniform normalized Euclidean bilinear-kernel mass, exact standard-six-torus Fourier realization, and the frequency-uniform periodic $O(\delta)$ bilinear endpoint bound are also proved. Exact finite-Fourier summation, inner-chart identification with the coherent PDE interaction, mass-preserving isotropic dyadic rescaling, exact four-transverse/two-axial scaling, square-dyadic chart identification, its periodized coefficient formula, its uniform C/N mass

bound, and uniform frozen oriented-frame transference, physical input/output conjugation, and the pointwise endpoint for an arbitrary frame field are now proved as well. Four-sign coverage of the full normalized annulus, exact partition assembly, one uniform inverse-Fourier mass bound for the assembled multiplier, its exact uncut coherent-symbol identity on the annulus, and its physical variable-frame C/N endpoint are also proved. The finite-support annular sector projector is now constructed, its six-dimensional chart is identified with the two independently filtered mode charts, and the genuine rotated radial-shell/angle-cone selector is proved to lie in that annulus. The exact physical coherent formula follows without an additional support premise. The pointwise strain-derived axial frame and its expanding-line orientation are also constructed. Finally, the full physical finite-sector stretching sum is proved to be exactly that annular coherent action plus the three-term misalignment remainder, with a uniform C/N bound on the first term, including in the strain-derived frame. The complex coefficient remainder is now bounded by transverse expanding-line energy, first modewise, then after finite-sector summation, and finally by an exact physical-space L^2 integral via finite Parseval, with no support-cardinality loss. A sign-invariant line-projector theorem controls the frozen-to-varying-frame error by $4\varepsilon/\text{threshold}$ times the full L^2 energy, with the ordinary frozen top-gap band returned as the alternative. On its separated branch, homogeneous spectral-gap coercivity now bounds the transverse energy by the integrated top Rayleigh defect. The separated-cell finite sum, generic bounded-overlap integration, the actual 125-cell weighted sum with cell-dependent frozen strains, and continuity/measurability of the sign-invariant projector on every positive-gap region are now checked. On the small-top-gap branch the bottom-separated expanding-plane projector is also measurable, while simultaneous collapse of both adjacent gaps gives a direct small stretching-rate bound for trace-free strain. The localization/commutator connection from global reconstructions to spatial cells is now checked through the concrete smooth cover. Exact finite cutoff convolution, Minkowski-sum support leakage, and the sharp physical L^2 cutoff bound are now proved, as is the exact buffered annulus containing every shifted output mode. A denominator-safe four-chart multiplier is now constructed on that buffer, proved exactly coherent there, and given one aperture-uniform inverse-kernel L^1 bound. Its complete square-dyadic, rotated, periodized, and physical transfer is proved, and finite-band cutoff products are connected to it with no support-cardinality loss. The actual spatial cutoff is now a periodized compactly supported smooth bump; its Fourier coefficients are proved absolutely summable by Schwartz decay and Poisson transference. Its explicit 125 translates cover the torus, admit a common scale-compatible finite approximation, and satisfy the uniform smooth-cell-to-coherent $O(\varepsilon/N)$ transfer. Complete ordinary dyadic shells for arbitrary Fourier data are also constructed, each is covered by two adjacent square-dyadic scales, and each has an exact four-way lower/upper and pancake/nonpancake split. For componentwise L^2 data these shell energies are summable and, together with the zero mode, equal the full physical energy.

The remaining field-level steps are the smooth-projector collar, dynamical coherence inside the bottom-separated expanding plane, aggregation of these branches over space and time, construction of the cellwise frozen strains from the actual strain field with the needed oscillation, conversion of the resulting L^2/defect estimate to the required scale-local endpoint budget, and assembly of the hard dyadic pancake split into `coherent_le` for the actual Navier–Stokes functionals. (ii) The misalignment budget itself: dynamical, time-integrated control of the misalignment share; the checked fences of Sections 2, 4.6 and the static no-go apply to it verbatim.

6 Opening the dynamical stage: direction rate and the tilting gap

The first dynamical audit separates two notions that should not be conflated: rotation of the vorticity direction and rotation of the strain eigenframe. Both are relevant to a pancake configuration, but they obey different equations.

For a nonzero vorticity vector written $\omega = r\xi$, with $|\xi| = 1$, and a material forcing F , define the transverse projection

$$P_\xi^\perp F = F - (\xi \cdot F)\xi, \quad \dot{\xi} = r^{-1}P_\xi^\perp F.$$

The module `DirectionEvolutionTilting` verifies this formula from a differentiable factorization $\omega = r\xi$, including tangency $\xi \cdot \dot{\xi} = 0$. If the pointwise vorticity equation is written as $F = rS\xi + d$, where S is the strain and d is the diffusive remainder, the exact split is

$$\dot{\xi} = P_\xi^\perp(S\xi) + r^{-1}P_\xi^\perp d.$$

For two differentiable direction paths, the same module proves

$$\frac{d}{dt}|\xi_1 - \xi_2|^2 = 2(\xi_1 - \xi_2) \cdot (\dot{\xi}_1 - \dot{\xi}_2),$$

together with its Cauchy–Schwarz bound by $2|\xi_1 - \xi_2|(|\dot{\xi}_1| + |\dot{\xi}_2|)$. Thus total direction turning is an exact dynamical input to misalignment production; obtaining the scale-critical time integral required by `MisalignmentStrainBudget` remains analytic work.

The module `PancakeDyadicDirectionEvolution` makes this kinematics apply to the same hard block used by the Fourier misalignment estimate. For the square-dyadic pancake selector $P_{N,F}$ at a fixed spatial point, it proves coefficientwise that

$$\partial_t \Re(P_{N,F}\omega) = \Re(P_{N,F}\dot{\omega}).$$

If this real block is $r\xi$, the exact direction equation is

$$\dot{\xi} = P_\xi^\perp(S\xi) + r^{-1}P_\xi^\perp R_{N,F}, \quad R_{N,F} = \Re(P_{N,F}\dot{\omega}) - rS\xi.$$

Thus the previously separate physical-direction and Fourier-block statements now meet at a single explicit residual. This is not yet a bound on that residual: its Navier–Stokes decomposition and scale-critical time estimate are the next dynamical obligations. For any proposed coefficient field G representing the stretching part, the same module further proves the exact split

$$R_{N,F} = \Re P_{N,F}(\dot{\omega} - G) + (\Re P_{N,F}G - rS\xi).$$

The first term is the projected equation remainder (transport and viscosity for the vorticity equation); the second is the projection/variable-strain defect once G is identified with the coefficients of $S\omega$. This separates the two estimates needed for closure instead of charging both to an undifferentiated direction-coherence premise. The companion module `PancakeFourierStretchingEvolution` constructs the canonical phase-suppressed coefficient field: every retained source/receiver pair is collected in the fiber $k + \ell = q$, and finite reconstruction of the resulting output coefficients is proved exactly equal to the previously defined pair-indexed complex stretching polynomial. The module `PancakePeriodicVorticityEquation` then fixes the unit-torus constants. Writing $D = 2\pi i$, it defines the physical Biot–Savart coefficient as $-D^{-1}|k|^{-2}k \times \hat{\omega}_k$ and proves exactly that physical strain stretching is the negative of the suppressed symbol. It also derives the right-hand-side transport sign, includes the multiplier $\nu D^2|q|^2$, and packages the finite coefficient equation. Under that

equation the first block residual is proved to be precisely projected transport plus viscosity; the second is the physical stretching projection/variable-strain defect. What remains is a scale-critical estimate for those two concrete channels and a controlled passage from finite convolutions to the full periodic solution, not a normalization ambiguity. The companion module **PancakeBlockReality** proves that both the radial shell and oriented squared-aperture predicate are invariant under frequency negation. Reindexing the finite Fourier sum by $k \mapsto -k$, using character and coefficient conjugate symmetry, proves $\overline{P_{N,F}\omega} = P_{N,F}\omega$ coordinatewise. It follows that the complex block has zero imaginary part and complexifying the real block used above returns the full reconstruction exactly. The same module then checks symbol-by-symbol conjugate covariance of the physical unit-torus equation and reindexes both convolution supports simultaneously. For negation-closed supports and real coefficient fields it proves $\text{RHS}(-q) = \overline{\text{RHS}(q)}$, so the finite dynamics is internally consistent with the real block on which the direction law is stated. The next exact cancellation is material transport. If B is the hard pancake mode set, then

$$P_B[-(u \cdot \nabla)\omega] + (u \cdot \nabla)(P_B\omega) = [u \cdot \nabla, P_B]\omega$$

is checked coefficientwise for finite supports. For an output $q \in B$ the surviving sum contains only receiver frequencies initially outside B (incoming transfer); for $q \notin B$ it is minus the sum from receivers inside B (outgoing transfer). Thus following the fluid removes bulk advection and leaves an explicit frequency-boundary flux.

The pressure Hessian enters through the strain-eigenframe equation instead. Differentiating $Se_i = \lambda_i e_i$ and pairing with e_j , $i \neq j$, gives

$$\langle e_j, \dot{e}_i \rangle = \frac{\langle e_j, \dot{S}e_i \rangle}{\lambda_i - \lambda_j}.$$

The anisotropic pressure Hessian contributes to the numerator. The checked theorem **no_uniformGapFreePressureTiltingBound** records the necessary denominator: no nonnegative constant bounds the angular rate from a bounded off-diagonal torque independently of the strain eigenvalue gap. Unit torque with gap $1/(C+1)$ already defeats a proposed constant C .

This rules out a raw pressure-Hessian norm as the next envelope. A viable dynamical continuation must instead control a gap-weighted pressure torque, or prove cancellation/residence-time control over near-degenerate intervals, and then transfer that control to the Littlewood–Paley misalignment share.

The same denominator is now proved for spatial frame freezing. In **PancakeEigenframePerturbation**, if $Ae = \lambda e$, $Bf = \mu f$, B is self-adjoint, and e, f are unit vectors, then

$$(\lambda - \mu)\langle f, e \rangle = \langle f, (A - B)e \rangle, \quad |\lambda - \mu| |\langle f, e \rangle| \leq \|A - B\|.$$

Consequently, at every positive threshold, either the eigenvalues are in the near-degenerate band or the frame overlap is bounded by the strain oscillation divided by that threshold. This is the exact gap/near-degeneracy alternative needed by the variable-frame remainder; it is not yet the scale-summed bound needed for closure.

Two further modules turn this into an intrinsic local dynamics statement. **PancakeStrainSpectralFrame** constructs, for every symmetric three-dimensional strain operator, an ordered orthonormal eigenframe and proves the three-coordinate Parseval and Rayleigh identities. Thus the spectral coordinates used below are constructed objects, not assumed data. For a unit vorticity direction ξ and a moving unit top eigenvector e , **PancakeAlignmentDynamics** defines

$$E(\xi, e) = 1 - \langle \xi, e \rangle^2$$

and proves the exact simple-gap identity

$$\dot{E} = -2\langle \xi, e \rangle^2 (\lambda_0 - \langle \xi, S\xi \rangle) - 2\langle \xi, e \rangle (\langle d, e \rangle + \langle \xi, \theta \rangle),$$

where d is the non-strain direction forcing and $\theta = \dot{e}$. The Rayleigh defect is bounded below by the top spectral gap times E , and the module integrates this damping to obtain the residence estimate

$$\int_0^T (\lambda_0 - \lambda_1) E dt \leq E(0) + 2 \int_0^T |\langle d, e \rangle + \langle \xi, \theta \rangle| dt$$

under its stated regularity and simple-gap hypotheses. It also proves the complementary upper bound by the full spectral width.

These identities expose rather than hide the hard cases. Substituting the usual eigenvector perturbation estimate $|\theta| \lesssim |\dot{S}|/(\lambda_0 - \lambda_1)$ into the residence inequality and applying a naive Young inequality produces still worse inverse-gap costs; it is not a closure. Moreover, a large trace-free strain may be isotropic on a strongly expanding two-plane. Such a partially degenerate plane supplies damping toward the plane but does not select a coherent line inside it. Controlling those intervals requires a projector/cluster formulation or a genuine time-integrated cancellation, and remains part of the dynamical pin.

7 Status and scope

We state the scope of this finding exactly.

What is established from the original audit. Hypothesis (H1) of v7 — a uniform $H^m \rightarrow H^m$ operator bound (3) on the adjoint / push-forward Ad_g over an H^m chart-neighborhood of the identity — is *false* in the normalized Fourier-mode shear model of Section 2. The refutation is machine-checked (no sorry, elementary) in `Mettapedia.FluidDynamics.NavierStokes.BenH1Break`, and the underlying norms are independently derived from the Sobolev-weight structure in the companion numeric lab. The model faithfully captures the mechanism that defeats v7’s local-boundedness argument: the adjoint (7) loses one derivative ($g \in H^m \Rightarrow Dg \in H^{m-1}$), so on a fixed-radius H^m chart the operator norm is unbounded along high-wavenumber shears.

What is not established. This is a refutation of (H1) *in a model* — a single analytic Fourier-mode family together with the linearized commutator action — not a from-Mathlib-Sobolev-first-principles theorem about $\text{SDiff}^m(\mathbb{T}^3)$. The model uses the exact push-forward formula (7) on an explicit shear family and the standard Fourier H^m norm; it does not formalize the full diffeomorphism-group functional analysis. The mechanism (derivative loss) is captured faithfully, but the strength of the conclusion is “checked in this model,” not “checked over the entire chart from Mathlib’s Sobolev theory.”

Consequence for v7. Because every quantitative step of the SG–Cole–Hopf program inherits the constant C_{Ad} from (3) — through (5), (2), and (1) — the falsity of (H1) as stated means the route does *not*, as written, establish global regularity. v7 reduces 3D Navier–Stokes global regularity to (H1) (equivalently, to a uniform curvature / Bakry–Émery-type bound), and that reduction target does not hold uniformly in the form required.

What the repair establishes. The stochastic-Lagrangian repair is not refuted by the H1 countermodel, and that evasion is itself a checked theorem (Section 4.1). Machine-checked without `sorry` across the campaign: the one-form calculus with the exact frame-Laplacian identity; the divergence-free non-exact witness forcing the gauge; the pushdown theorem producing Navier–Stokes with the explicit pressure $p = \partial_t W + \frac{1}{2}\langle u, u \rangle + \langle u, \nabla W \rangle - \nu \Delta W$ from the averaged transport hypothesis; the gate consumers and the assembled conditional route; nonzero exact inhabitants (decaying shear with balanced gauge) and end-to-end rest canaries for both the coarse and the refined route; the frozen-strain exact solution with its unbounded-amplification and coverage-gap theorems; the aperture dichotomy–budget equivalence with both gap-band failure theorems and the Grönwall-saturation bound; the single-mode Biot–Savart null structure; and the dyadic absorption algebra with explicit constants. The theorem `stochasticLagrangian_conditional_route_of_pancakeBudget` records that the route closes if the scale-local pancake strain budget and the non-pancake sector estimates are supplied.

Scope of the new pieces. The same honesty discipline as for the H1 model applies. The frozen-strain results are theorems about a linear prescribed-strain model, labeled as such; the aperture theorems are exact inequalities between the model’s threshold parameters; the plane-wave null structure is an identity for one exact Fourier mode (sector aggregation is exactly the open budget); the averaged transport equation enters the pushdown as a hypothesis field (no stochastic calculus is formalized); and the coercive/residual sector estimates are named interfaces consumed from the source analysis, not re-proved. None of these statements is a claim about general Navier–Stokes solutions beyond what its label says.

Verification metadata. The named campaign modules contain no `sorry`, `admit`, or unsafe definitions. Selected load-bearing theorems carry explicit machine axiom audits; the remaining newly added headlines, including the finite Parseval misalignment-energy bridge, are checked by the Navier–Stokes live surface and the same explicit audit boundary. The live surface builds green. Earlier conditional route theorems retain their instantiation canaries demonstrating that their stated hypotheses are simultaneously satisfiable. This is a claim about proof integrity, not about discharge of the explicitly open analytic hypotheses.

What remains open. We make no claim to have proved the Millennium Navier–Stokes statement. The latest pancake-closure proposal fails as a complete proof because its aperture split requires the pointwise scale-local condition $\gamma \leq \nu \lambda_q$, and that condition is exactly the burden of the unresolved self-consistent strain estimate. After the September 2026 refinement (Section 5), the open analytic input is `MisalignmentStrainBudget` together with the dynamical identification and aggregation of the now-concrete hard dyadic coherent/misaligned transfer, which jointly reproduce `ScaleLocalPancakeStrainBudget` (cross-referenced with the coarser `StochasticStretchingEstimate`). A successful completion must control dynamical coherence inside the stable expanding-plane part of the collision band (the separated line/plane projectors are measurable and the triple-collision strain is small), pass from the hard selector to the intended smooth projector, identify the now-exact hard-block physical split with the route’s scale-local misalignment functional, aggregate the proved frozen-cell/small-gap projector alternative and convert its L^2 control to the endpoint scale budget, and prove a dynamical, time-integrated control of the vorticity-direction misalignment share, strong enough to cover the gap band identified above. The finite four-sign annular partition and its exact periodic physical transference are proved; the remaining identification may not assign the aperture gain to the projection. Section 6 further shows

that a pressure-Hessian route must carry inverse strain-eigenvalue-gap information or an equivalent time-integrated cancellation; pressure-Hessian size alone cannot close the pin. Once the analytic estimates are closed, the remaining endpoint work is to promote the common pointwise majorant to the measurable spatial essential-supremum BKM integrand and apply a formal continuation theorem for the chosen solution class.

Provenance

The manuscript discussed is Ben Goertzel, *Global Regularity of 3D Navier–Stokes via SG–Cole–Hopf Program*, version v7, together with the subsequent pancake-closure repair proposal audited in July 2026. The formalization, Fourier-mode model, repaired conditional route, pancake adjudication, and numeric lab are work of the MeTTapedia formalization effort. The repaired-route development comprises, under Mettapedia.FluidDynamics.NavierStokes: BenH1Break, OneFormFrameCalculus, OneFormExactness, GradientOneForm, FrozenStrainModel, ApertureCoverageDichotomy, PlaneWaveSelfStrain, and the StochasticLagrangian layer (PullbackOneForm, ShearJacobianBounds, ConstantinIyerRepresentation, VorticityStretchingGate, DecayingShearInstance, DyadicPancakeClosure, PancakeCoherentPairEstimate, PancakeRealCoherentPairEstimate, PancakeConeOperatorKernel, PancakePeriodicCoherentSplit, PancakePeriodicComplexStretch, PancakePeriodicOperatorKernel, PancakeFrameCovariance, PancakeVariableFrameCommutator, and PancakeEigenframePerturbation, PancakeAlignmentDynamics, PancakeStrainSpectralFrame, PancakeSmoothCoherentSymbol, PancakeSmoothCoherentAperture, PancakeCoherentStretchFactorization, PancakeSmoothCoherentStretchSymbol, and PancakeBilinearKernelScaling, and PancakeUniformKernelFamily, and PancakeBilinearPeriodization, PancakeFiniteFourierBilinearTransfer, PancakeDyadicKernelScaling, PancakeSquareDyadicFiniteTransfer, and PancakeFrozenFrameKernelTransfer, and PancakePhysicalFrameKernelTransfer, through the later annular, full-strain-frame, coercivity, cell-aggregation, finite-localization, and PancakeBufferedAnnularMultiplier, buffered physical-transfer, and PancakeLocalizedBufferedTransfer, bilinear stability, and buffered localization-stability, smooth-cutoff Fourier-tail, PancakeConcreteSmoothCellCutoff, PancakeConcreteSpatialCoverTransfer, and PancakeArbitraryDataBlocks and PancakeArbitraryDataMisalignment, and PancakeConcreteCoverEnergy, PancakeConcreteWeightedFreezing, and PancakeSpectralProjectorRegularity, and PancakeSpectralClusterDichotomy, and PancakeExpandingPlaneResidence, and PancakeDyadicDirectionEvolution, and PancakeFourierStretchingEvolution, and PancakePeriodicVorticityEquation, and PancakeBlockReality, and PancakeMaterialTransportCommutator modules), each with a regression companion where applicable. The aperture module is retained as an explicitly attenuated analytic model; the exact coherent projection no-go above prevents its use as the desired field-transfer estimate. Background references: J. T. Beale, T. Kato, A. Majda, *Comm. Math. Phys.* **94** (1984), 61–66; P. Constantin, G. Iyer, *Comm. Pure Appl. Math.* **61** (2008), 330–345; D. Ebin, J. Marsden, *Ann. of Math.* **92** (1970), 102–163; H. Bahouri, J.-Y. Chemin, R. Danchin, *Fourier Analysis and Nonlinear Partial Differential Equations*, Springer, 2011; P. Constantin, C. Fefferman, *Indiana Univ. Math. J.* **42** (1993), 775–789; J. Tom, M. Carbone, A. D. Bragg, *J. Fluid Mech.* **910** (2021), A24.